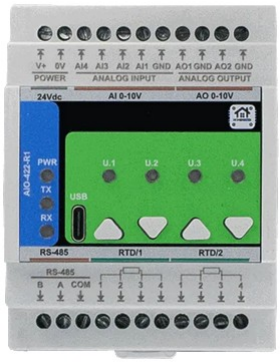
Datasheet

AIO-422-R1 Analog I/O Module



Part No.: AIO-422-R1

Manufacturer: ISYSTEMS AUTOMATION S.R.L.

Hardware Version: V1.0

Architecture: Modular (MCU Board + Field Board)

Controller: RP2350A

Functional Overview

The AIO-422-R1 is a DIN-rail mounted mixed-signal expansion module providing 4 analog voltage inputs (0–10 V), 2 analog voltage outputs (0–10 V), and 2 RTD temperature channels. The module communicates via RS-485 (Modbus RTU) and supports USB-C configuration and firmware update.

Board	Function
MCU Board	RP2350A controller, RS-485 communication, USB-C interface, power conversion
Field Board	4x Analog Inputs, 2x Analog Outputs, 2x RTD inputs

The module operates as a Modbus RTU slave over RS-485 and supports configuration and firmware update via USB-C.

Technical Specifications

Specification	Details
Microcontroller	RP2350A dual-core microcontroller
Storage	External QSPI Flash (32 Mbit)
Power Input	24 V DC nominal
Input Protection	1 A fuse, reverse polarity diode, TVS surge suppression, EMI filtering
Main Logic Supply	Buck regulator 24 V → 5 V, 3.3 V LDO regulator
Document revision	DS-AIO-422-R1 Rev. B · 2026-09 · Hardware R1 (V1.0)
Analog Inputs	4 × voltage inputs, 0–10 V nominal range
ADC	ADS1115, 16-bit resolution
Input Impedance	>100 kΩ
Analog Outputs	2 × voltage outputs, 0–10 V nominal range
DAC	MCP4725A0, 12-bit resolution
Max Output Current	10 mA recommended per channel
RTD Channels	2 × PT100/PT1000 (2/3/4-wire)
RTD Interface	MAX31865, 15-bit temperature resolution
User Interface	4 buttons; 9 LEDs (power, RX, TX, 4 user; 2 unassigned — MCU board shared with ENM-223-R1 / WLD-521-R1, where those two indicate relays)
RS-485	half-duplex Modbus RTU, not galvanically isolated
USB	USB-C, 5 V logic, ESD protected
Typical Power Consumption	0.2–0.4 W typical, approx. 1 W maximum (logic only)
Modbus defaults	Address 3, 19200 baud, 8N1

Typical module power consumption: 0.2–0.4 W typical, approx. 1 W maximum (logic only).

Top Terminal Block (1–11)

Front view – terminals numbered from left to right.

Pos	Markin g	Signal	Description

1	V+	POWER	24 V DC input
2	0V	POWER	24 V DC return
3	AI4	ANALOG_INPUT	Analog input 4, 0-10 V
4	AI3	ANALOG_INPUT	Analog input 3, 0-10 V
5	AI2	ANALOG_INPUT	Analog input 2, 0-10 V
6	AI1	ANALOG_INPUT	Analog input 1, 0-10 V
7	GND	ANALOG_INPUT	Common return for AI1-AI4
8	AO1	ANALOG_OUTPUT	Analog output 1, 0-10 V
9	GND	ANALOG_OUTPUT	AO1 return
10	AO2	ANALOG_OUTPUT	Analog output 2, 0-10 V
11	GND	ANALOG_OUTPUT	AO2 return

Bottom Terminal Block (1–11)

Front view – terminals numbered from left to right.

Pos	Marking	Signal	Description
1	B	RS485	RS-485 data -
2	A	RS485	RS-485 data +
3	COM	RS485	RS-485 signal reference
4	1	RTD1	RTD 1 terminal 1
5	2	RTD1	RTD 1 terminal 2
6	3	RTD1	RTD 1 terminal 3
7	4	RTD1	RTD 1 terminal 4
8	1	RTD2	RTD 2 terminal 1
9	2	RTD2	RTD 2 terminal 2
10	3	RTD2	RTD 2 terminal 3
11	4	RTD2	RTD 2 terminal 4

Installation, Environmental & Mechanical

Specification	Details
Terminal Type	Pluggable screw terminal blocks, 5.08 mm pitch
Wire Cross Section	0.2–2.5 mm^2 (AWG 24–12)
Tightening Torque	0.4–0.6 Nm

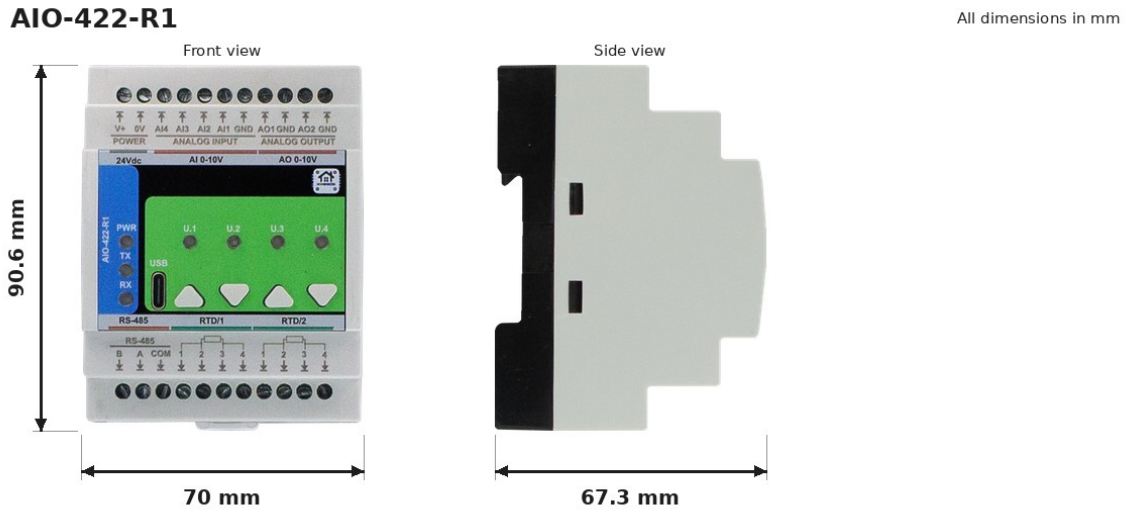
Specification	Details
Operating temperature	0 °C to +40 °C
Storage temperature	−10 °C to +55 °C
Relative humidity	0–90 % RH, non-condensing
Ingress protection	IP20 (inside cabinet only)
Installation	Indoor control cabinet only; not for outdoor or exposed installation
Maximum altitude	2000 m

Pollution degree	2

Specification	Details
Dimensions	70 × 90.6 × 67.3 mm (L × W × H)
DIN width	4 modules (≈ 70 mm)
Mounting	35 mm DIN rail
Enclosure	PC/ABS industrial enclosure
Terminal type	Pluggable screw terminal blocks, 5.08 mm pitch
Wire cross-section	0.2–2.5 mm² (AWG 24–12)
Tightening torque	0.4–0.6 Nm
Net weight	160 g
Gross weight	280 g
Pack size	140 × 125 × 94 mm (L × W × H)
Mechanical drawing	Mechanical drawing: front and side view, dimensions in mm
Notes	All dimensions shown in millimeters unless stated otherwise

Install only inside a control cabinet with ventilation; the cabinet must include a protective front plate covering all module connection terminals and a closing protective door; not for outdoor or exposed installation.

All wiring terminals must be protected against accidental contact by an insulating front plate, wiring duct, or terminal cover. Exposed live terminals are not permitted.



Cable Recommendations & Shield Grounding

This section applies to Analog (0–10V), Temperature (RTD), and RS-485. Use shielded, twisted constructions and bond shields correctly to reduce EMI and ground-loop issues.

General Routing Rules

- Route low-level signal cables separately from power wiring.
- If crossing power cables is unavoidable, cross at 90°.
- Keep cable runs as short as practical.
- Avoid parallel runs with high-current conductors.

Analog (0–10V) Cable

- Construction: twisted pair (Signal + GND) per channel
- Shielding: overall shield (standard) or individually shielded pairs (high-EMI)

Examples: J-Y(ST)Y (overall shield) or LI2YCY PIMF (shielded twisted pairs; one pair per channel)

Temperature Cable

RTD (PT100/PT1000)

Recommended: shielded multi-core for 2/3-wire; shielded pairs for best accuracy (4-wire)

Examples: J-Y(ST)Y (overall shield) or LI2YCY PIMF (pairs; e.g. 2×2×0.50 for 4-wire)

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RS-485 Cable

- Twisted pair recommended (120 Ω characteristic impedance).
- Shielded cable recommended in industrial environments.
- Use one twisted pair for A/B.

- Run COM to every node. Required, not optional: the RS-485 port is not galvanically isolated.

Examples: J-Y(ST)Y 2×2×0.5 mm² or LI2YCY PiMF 2×2×0.50

Shield Grounding

- Bond cable shield(s) to cabinet PE/EMC ground at one end (PLC side recommended).
- Do not connect shields directly to signal terminals (A/B/COM).

Compliance & Certifications

The AIO-422-R1 module is CE marked and designed to comply with applicable European Union directives. The manufacturer maintains technical documentation and a signed EU Declaration of Conformity (DoC).

EMC 2014/30/EU · LVD 2014/35/EU · RoHS 2011/65/EU

Standards: EN 61000-6-1 · EN 61000-6-3 · EN 62368-1 · EN IEC 63000

Safety Notice

The power input and signal terminals are SELV circuits.

Installation by qualified personnel only.

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